



CH 4

Digital Logic Design

Midterm

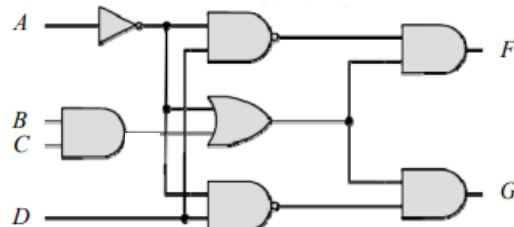
2025/04/28

姓名：

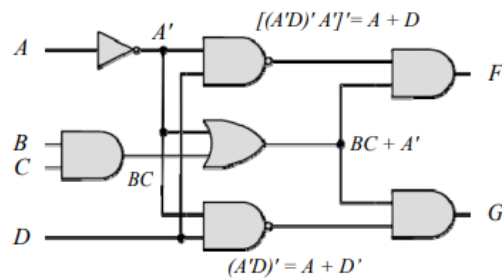
學號：

系級：

- Obtain the simplified Boolean expressions for output F and G in terms of the input variables in the following circuit. (10%)



4.2



$$F = (A + D)(A' + BC) = A'D + ABC + BCD = A'D + ABC$$

$$F = (A + D')(A' + BC) = A'D' + ABC + BCD' = A'D' + ABC$$

AB \ CD		C			
		00	01	11	10
A	00	m_0	m_1 1	m_2 1	m_3
	01	m_4	m_5 1	m_6 1	m_7
	11	m_{12}	m_{13}	m_{15} 1	m_{14} 1
	10	m_8	m_9	m_{11}	m_{10}
		D			

$$F = A'D + ABC + BCD = A'D + ABC$$

AB \ CD		C			
		00	01	11	10
A	00	m_0 1	m_1	m_2	m_3 1
	01	m_4 1	m_5	m_6	m_7 1
	11	m_{12}	m_{13}	m_{15} 1	m_{14} 1
	10	m_8	m_9	m_{11}	m_{10}
		D			

$$G = A'D' + ABC + BCD' = A'D' + ABC$$

$$F=G$$

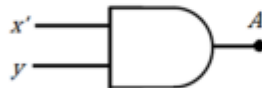
2. Design a combinational circuit with three inputs X , Y and Z , and three outputs A , B and C . When the binary input is 0, 1, 2, or 3, the binary output is twice the input. When the binary input is 4, 5, 6, or 7, the binary output is half the input. (20%)

4.5

x	y	z	A	B	C
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	0	1	1
1	1	1	0	1	1

		A			
				y	
yz	x	00	01	11	10
x	0	m_0 0	m_1 0	m_3 1	m_2 1
	1	m_4 0	m_5 0	m_7 0	m_6 0

$$A = x'y$$



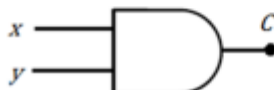
		B			
				y	
yz	x	00	01	11	10
x	0	m_0 0	m_1 1	m_3 1	m_2 0
	1	m_4 1	m_5 1	m_7 1	m_6 1

$$B = x + z$$



		C			
				y	
yz	x	00	01	11	10
x	0	m_0 0	m_1 0	m_3 0	m_2 0
	1	m_4 0	m_5 0	m_7 1	m_6 1

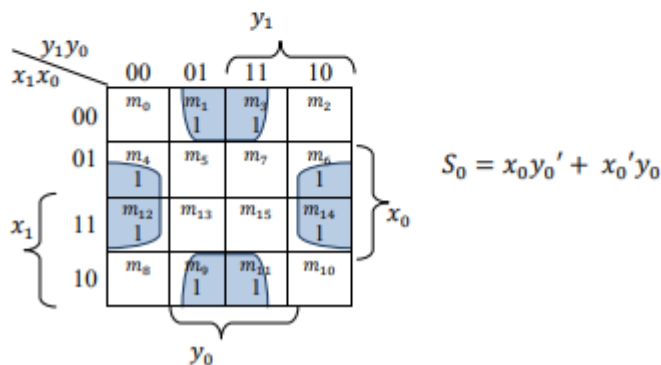
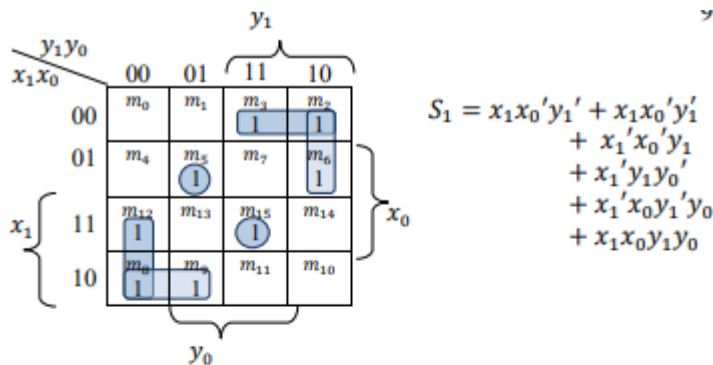
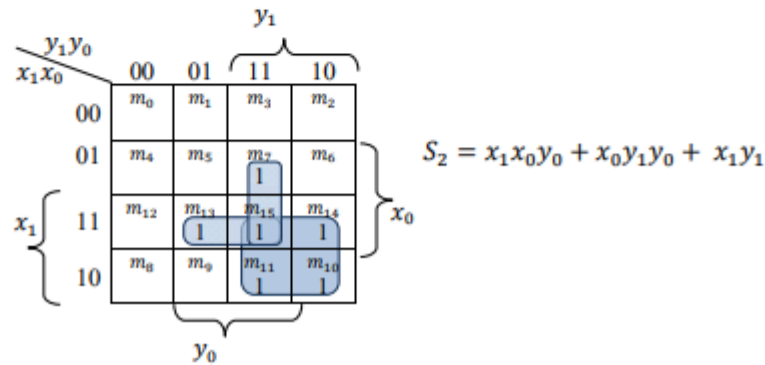
$$C = xy$$



3. Design a combinational circuit that adds two 2-bit numbers. The circuit outputs a three bit number that represents the arithmetic sum of the inputs. (15%)

4.21

x_1	x_0	y_1	y_0	S_2	S_1	S_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0



4. Design a BCD-to-decimal decoder using the unused combinations of the BCD code as don't-care conditions. (10%)

4.24

Inputs: A, B, C, D

$$D_0 = A'B'C'D'$$

$$D_1 = A'B'C'D$$

$$D_2 = B'CD'$$

$$D_3 = B'CD$$

$$D_4 = BC'D'$$

Outputs: $D_0, D_1, \dots, D_9$

$$D_5 = BC'D$$

$$D_6 = BCD'$$

$$D_7 = BCD$$

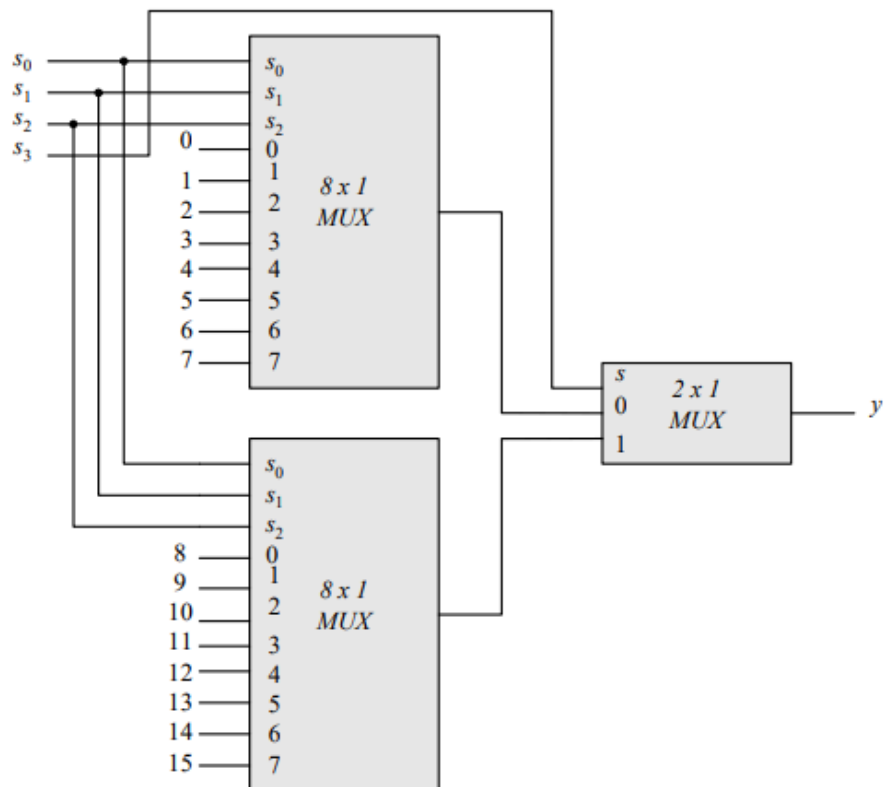
$$D_8 = AD'$$

$$D_9 = AD$$

AB		CD			
		00	01	11	10
A	00	m_0 D_0	m_1 D_1	m_3 D_3	x D_2
	01	m_4 D_4	m_5 D_5	m_7 D_7	m_6 D_6
	11	m_{12} x	m_{13} x	m_{15} x	m_{14} x
	10	m_8 D_8	m_9 D_9	m_{11} x	m_{10} x

5. Construct a 16×1 multiplexer with two 8×1 and one 2×1 multiplexers. Use block diagrams. (10%)

4.31

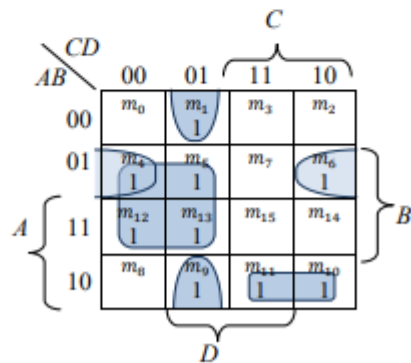


6. An 8×1 multiplexer has inputs A , B , and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs I_0 through I_7 are as follows: (20%)
 $I_1 = I_7 = 0$; $I_2 = I_5 = I_6 = 1$; $I_0 = I_4 = D$; and $I_3 = D'$.

4.34 (a)

A	B	C	D	I	F
0	0	0	0	$I_0 = D$	0
0	0	0	1		1
0	0	1	0	$I_1 = 0$	0
0	0	1	1		0
0	1	0	0	$I_2 = 1$	1
0	1	0	1		1
0	1	1	0	$I_3 = D'$	1
0	1	1	1		0
1	0	0	0	$I_4 = D$	0
1	0	0	1		1
1	0	1	0	$I_5 = 1$	1
1	0	1	1		1
1	1	0	0	$I_6 = 1$	1
1	1	0	1		1
1	1	1	0	$I_7 = 0$	0
1	1	1	1		0

$$F = \Sigma(1,4,5,6,9,10,11,12,13)$$



$$F = A'BD' + AB'C + B'C'D + BC'$$

7. Using a decoder and external gates, design the combinational circuit defined by the following three Boolean functions: (15%)

$$F_1 = x'y' + xz$$

$$F_2 = xy'z' + yz$$

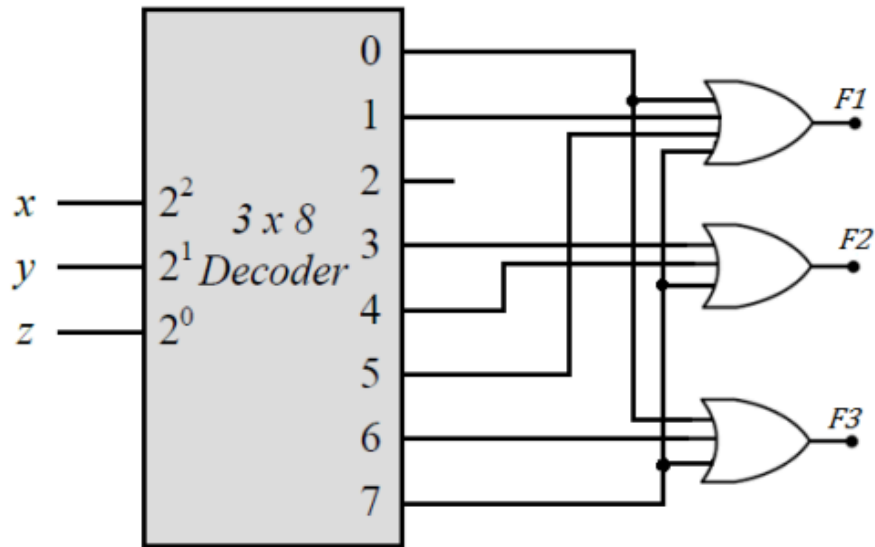
$$F_3 = x'y'z' + xy$$

4.28 (a)

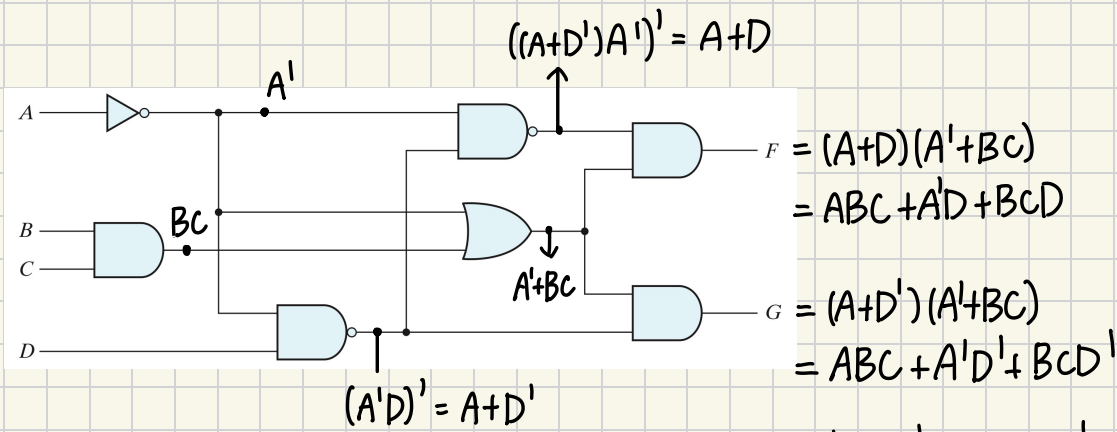
$$F_1 = x'y' + xz = x'y'z' + x'y'z + xy'z + xyz = \Sigma(0,1,5,7)$$

$$F_2 = xy'z' + yz = xy'z' + x'yz + xyz = \Sigma(3,4,7)$$

$$F_3 = x'y'z' + xy = x'y'z' + xyz' + xyz = \Sigma(0,6,7)$$



1.



$$F = ABCx + A'xxD + xBCD$$

111x 0xx1 x111

$$G = ABCx + A'xxD' + xBCD'$$

111x 0xx0 + x110

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	0	1	1
10	0	0	0	0

$$F = A'D + ABC$$

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	0	1	1
10	0	0	0	0

$$G = A'D' + ABC$$

2.

x	y	z	A	B	C
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	0	1	1
1	1	1	0	1	1

x \ yz	00	01	11	10
0	0	0	1	1
1	0	0	0	0

$$A = x'y$$

x \ yz	00	01	11	10
0	0	1	1	0
1	1	1	1	1

$$B = x+z$$

x \ yz	00	01	11	10
0	0	0	0	0
1	0	0	1	1

$$C = xz$$

3.

x_0	x_1	y_0	y_1	S_2	S_1	S_0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	1	0
0	1	0	1	0	1	1
0	1	1	0	1	0	0
0	1	1	1	1	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	1
1	0	1	0	1	1	0
1	0	1	1	1	1	1
1	1	0	0	1	1	0
1	1	0	1	1	1	1
1	1	1	0	1	0	0
1	1	1	1	1	0	1

$x_0 x_1 \backslash y_0 y_1$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	0	1	1
10	0	0	1	1

$$S_2 = x_0 x_1 y_1 + x_1 y_1 y_0 + x_0 y_0$$

$x_0 x_1 \backslash y_0 y_1$	00	01	11	10
00	0	0	1	1
01	0	1	0	1
11	1	0	1	0
10	1	1	0	0

$$S_1 = x_0' x_1' y_0' + x_0' y_0 y_1' + y_0' y_1' x_0 + x_0 x_1' y_0' + x_0' x_1 y_0' y_1 + x_0 x_1 y_0 y_1$$

$x_0 x_1 \backslash y_0 y_1$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

$$S_0 = x_1' y_1 + x_1 y_1'$$

4.

<key: BCD 0000 ~ 1001 10 ↑>

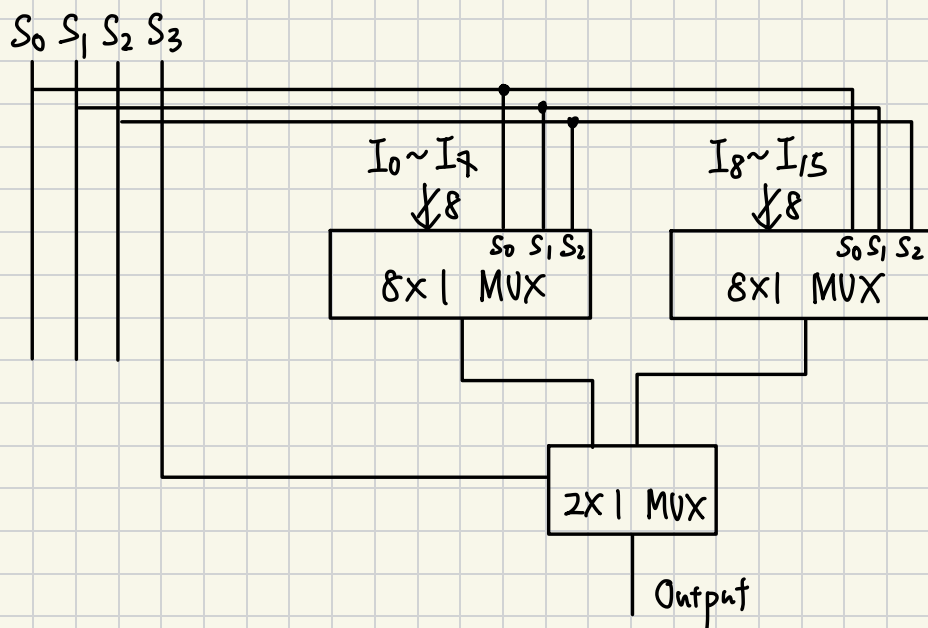
A	B	C	D	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉
0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1

AB \ CD	00	01	11	10
00	D₀	D₁	D₂	D₃
01	D ₄	D ₅	D ₇	D ₆
11	X	X	X	X
10	D ₈	D ₉	X	X

$$\begin{aligned} D_0 &= A'B'C'D' & D_5 &= BC'D \\ D_1 &= A'B'C'D & D_6 &= BCD' \\ D_2 &= B'CD' & D_7 &= BCD \\ D_3 &= B'CD & D_8 &= AD' \\ D_4 &= BC'D' & D_9 &= AD \end{aligned}$$

5.

$16 \times 1 \text{ MUX} : 16 \text{ IN, } 2^4 = 4 \text{ Selection Line}$



6.

A	B	C	D	I	Output
0	0	0	0	$I_0 = D$	0 0
0	0	0	1		1 1
0	0	1	0	$I_1 = 0$	0 2
0	0	1	1		0 3
0	1	0	0	$I_2 = 1$	1 4
0	1	0	1		1 5
0	1	1	0	$I_3 = D'$	1 6
0	1	1	1		0 7
1	0	0	0	$I_4 = D$	0 8
1	0	0	1		1 9
1	0	1	0	$I_5 = 1$	1 10
1	0	1	1		1 11
1	1	0	0	$I_6 = 1$	1 12
1	1	0	1		1 13
1	1	1	0	$I_7 = 0$	0
1	1	1	1		0

$$F = \sum(1, 4, 5, 6, 9, 10, 11, 12, 13)$$

AB \ CD	00	01	11	10
00	0	1	0	0
01	1	1	0	1
11	1	1	0	0
10	0	1	1	1

$$F = A'BD' + AB'C + C'D + BC'$$

7

$$F_1 = x'y' + xz$$

0 0 x | x |

$x \backslash yz$	00	01	11	10
0	1	1		
1		1	1	

$$F_1 = \Sigma(0, 1, 5, 7)$$

$$F_2 = xy'z' + yz$$

1 0 0 x | 1

$x \backslash yz$	00	01	11	10
0			1	1
1	1	1	1	

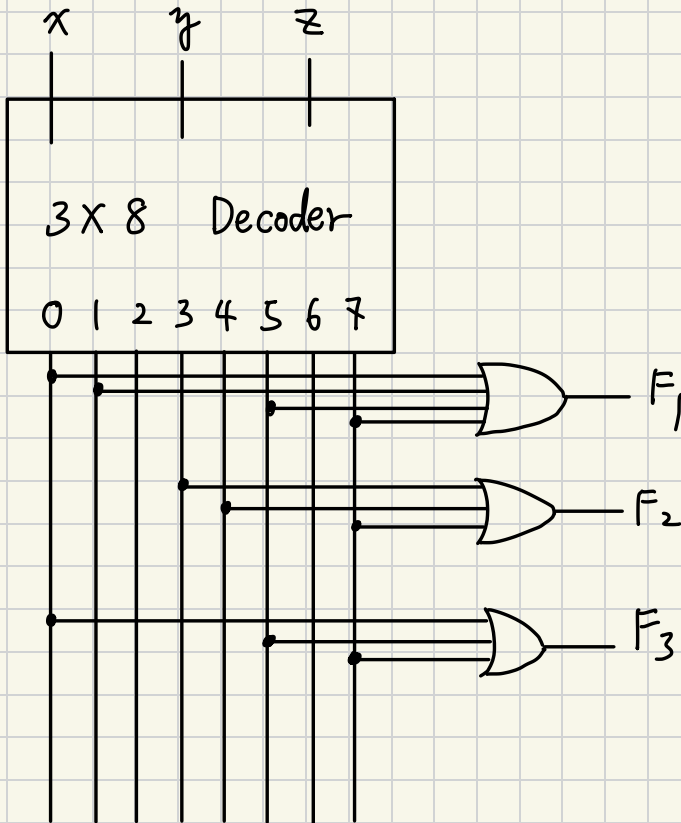
$$F_2 = \Sigma(3, 4, 7)$$

$$F_3 = x'y'z' + xy$$

0 0 0 | 1 x

$x \backslash yz$	00	01	11	10
0	1			
1			1	1

$$F_3 = \Sigma(0, 6, 7)$$



CH5

Digital Logic Design

Quiz 04

2025/05/19

姓名:

學號:

系級:

1. Show that the characteristic equation for the complement output of a T flip-flop is $Q(t+1) = TQ' + T'Q$ (20%)

5.3 $Q'(t+1) = (TQ' + T'Q)' = (TQ')' \cdot (T'Q)' = (T' + Q)(T + Q') = TQ + T'Q'$

2. A certain flip-flop has four operations: complement, set to 1, set to 0, and no change, when inputs A and B are 00, 01, 10, and 11, respectively. (20%)
 - (a) Tabulate the characteristic table.
 - (b) Tabulate the excitation table.
 - (c) Derive the characteristic equation.
 - (d) Show how the flip-flop can be converted to a T flip-flop.

5.4 (a) Characteristic Table:

A	B	Q(t+1)
0	0	Q'(t)
0	1	1
1	0	0
1	1	Q(t)

(b) Excitation Table

Q(t)	Q(t+1)	A	B
0	0	1	X
0	1	0	X
1	0	X	0
1	1	X	1

(c) Derivation of characteristic equation:

A	B	Q(t)	Q(t+1)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

		B			
		00	01	11	10
A	0	m ₀ 1	m ₁ 0	m ₃ 1	m ₂ 1
	1	m ₄ 0	m ₅ 0	m ₇ 1	m ₆ 0
		Q(t)			

$$Q(t+1) = A'Q' + BQ$$

(d) Connect A and B together and apply complement of T input to it.

3. A sequential circuit has two JK flip-flops A and B and one input x . The circuit is described by the following flip-flop input equations:(10%)

$$J_a = x' \quad K_a = B'$$

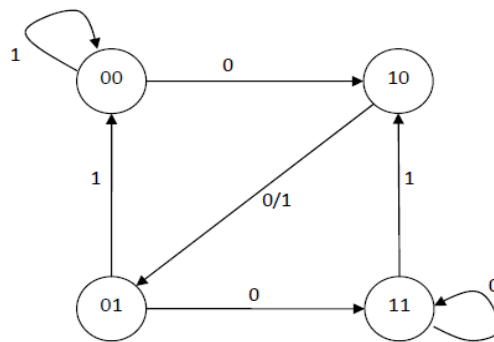
$$J_\beta = A \quad K_\beta = x$$

- (a) Derive the state equations $A(t+1)$ and $B(t+1)$ by substituting the input equations for the J and K variables.
 (b) Draw the state table and state diagram of the circuit.

5.9 (a) $J_A = x', K_A = B'; A(t+1) = JA' + K'A = x'A' + BA'$
 $J_B = A, K_B = x; B(t+1) = JB' + K'B = AB' + x'B$

(b)

Present State		Input	Next state	
$A(t)$	$B(t)$	x	$A(t+1)$	$B(t+1)$
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0



4. Starting from state a , and the input sequence 01110010011, determine the output sequence for
 (a) The state table of the previous problem.
 (b) The reduced state table from the previous problem. Show that the same output sequence is obtained for both.(20%)

5.13 Equivalent

(a) State: $a f b c e d g h g g h a$
 Input: $0 1 1 1 0 0 1 0 0 1 1$
 Output: $0 1 0 0 0 1 1 1 0 1 0$

(b) State: $a f b a b d g d g g d a$
 Input: $0 1 1 1 0 0 1 0 0 1 1$
 Output: $0 1 0 0 0 1 1 1 0 1 0$
 Equivalent states: $b = e, a = c, h = d$

5. Design a sequential circuit with two D flip-flops A and B , and one input x_{in} . (20%)
- (a) When $x_{in} = 0$, the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from 00 to 01, to 11, to 10, back to 00, and repeats.
- (b) When $x_{in} = 0$, the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from 00 to 11, to 01, to 10, back to 00, and repeats.

5.16 (a) $D_A = Ax' + Bx$

$$D_B = A'x + Bx'$$

Present state		Input x	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0

		B			
		x			
A	Bx	00	01	11	10
		m_0	m_1	m_3	m_2
0	0			1	
0	1	1		1	1
1	0				
1	1				

$$D_A = Ax' + Bx$$

		B			
		x			
A	Bx	00	01	11	10
		m_0	m_1	m_3	m_2
0	0		1	1	1
0	1				
1	0				
1	1				1

$$D_B = A'x + Bx'$$

(b) $D_A = A'x + Ax'$

$$D_B = AB + Bx'$$

Present state		Input x	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	1

		B			
		x			
A	Bx	00	01	11	10
		m_0	m_1	m_3	m_2
0	0			1	
0	1		1	1	
1	0	1			
1	1				1

$$D_A = A'x + Ax'$$

		B			
		x			
A	Bx	00	01	11	10
		m_0	m_1	m_3	m_2
0	0		1		1
0	1				
1	0			1	
1	1				1

$$D_B = AB + \overline{X}B + X\overline{AB}$$

1.

$$Q(t+1) = (TQ' + T'Q)$$

$$Q'(t+1) = (TQ' + T'Q)' = (TQ')'(T'Q)' = (T' + Q)(T + Q') = TQ + T'Q'$$

2.

(a)

A	B	$Q(t+1)$
0	0	$Q'(t)$ Comp'
0	1	1 Set
1	0	0 Set
1	1	$Q(t)$ No Change

(b)

$Q(t)$	$Q(t+1)$	A	B
0	0	1	X
0	1	0	X
1	0	X	0
1	1	X	1

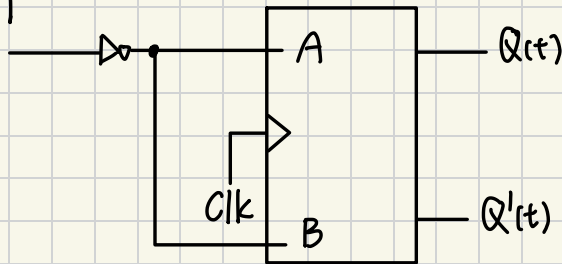
(c)

A	B	$Q(t)$	$Q(t+1)$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(d)

TFF 特徵表

T	$Q(t+1)$
0	$Q(t)$ No Change
1	$Q'(t)$ Comp'



A \ B Q(t)	00	01	11	10
0	1	0	1	1
1	0	0	1	0

$$Q(t+1) = A'Q' + BQ$$

3.

JK 特徵式: $JQ' + K'Q$

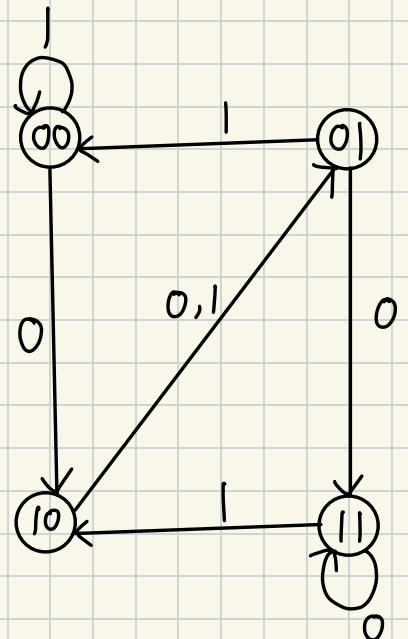
(a)

$$A(t+1) = A'A' + BA$$

$$B(t+1) = AB' + A'B$$

(b)

PS Input			NS	
A	B	X	A	B
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0



4.

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	f	b	0	0
b	d	a ^a	0	0
c	f	b ^b	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h ^d	0	1
h	g	a	1	0

P S	N S		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

IP SQ: 0111 0010 011

(a)

PS	a	f	b	c	e	d	g	h	g	g	h
IP	0	1	1	1	0	0	1	0	0	1	1
OP	0	1	0	0	0	1	1	1	0	1	0
NS	f	b	c	e	d	g	h	g	g	h	a

→ 0100 0111 010

(b)

PS	a	f	b	a	b	d	g	d	g	g	d
IP	0	1	1	1	0	0	1	0	0	1	1
OP	0	1	0	0	0	1	1	1	0	1	0
NS	f	b	a	b	d	g	d	g	g	d	a

→ 0100 0111 010

The (a) . (b) Output Sequence is same.

5.

(a) PS		IP	NS	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0

x=1 00 → 01 → 11 → 10

A \ Bx	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$A_{(t+1)} = Ax' + Bx$

A \ Bx	00	01	11	10
0	0	1	1	1
1	0	0	0	1

$B_{(t+1)} = A'x + Bx'$

5. (b)

PS			IP			NS		
A	B	X	A	B	X	A	B	X
0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	1	0
0	1	1	1	0	0	1	0	0
1	0	0	1	0	0	1	0	0
1	0	1	0	0	1	0	0	1
1	1	0	1	1	1	1	1	1
1	1	1	0	1	0	0	1	0

$X = 1 \Rightarrow 00 \rightarrow 11 \rightarrow 01 \rightarrow 10$

A \ Bx	00	01	11	10
0	0	1	1	0
1	1	0	0	1

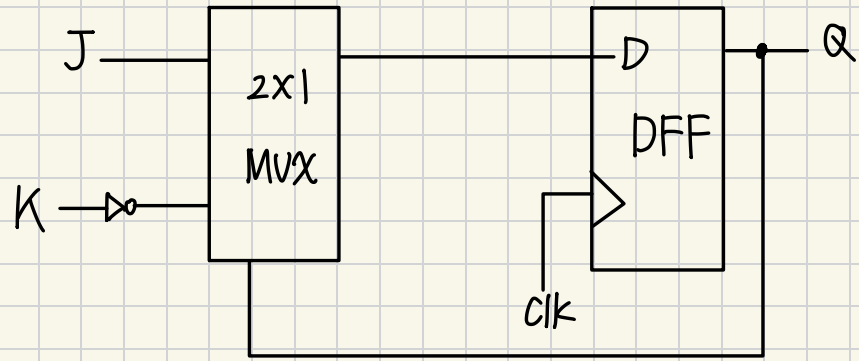
$A(t+1) = A'x + Ax'$

A \ Bx	00	01	11	10
0	0	1	0	1
1	0	0	1	1

$B(t+1) = AB + Bx' + A'B'x$

6. Construct a JK flip-flop using a D flip-flop, a two-to-one-line MUX, and an inverter.

$D = JQ' + K'Q$



5.4 Storage Element: Flip-Flops (12/16)

Other Flip-Flops

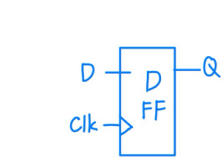


- Characteristic Tables (特徵表, 特性表)

Table 5.1
Flip-Flop Characteristic Tables

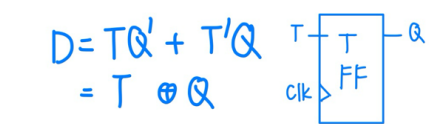
$D = JQ' + K'Q$

JK Flip-Flop			
J	K	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement



D Flip-Flop		
D	$Q(t+1)$	
0	0	Reset
1	1	Set

$D = TQ' + T'Q = T \oplus Q$



T Flip-Flop		
T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

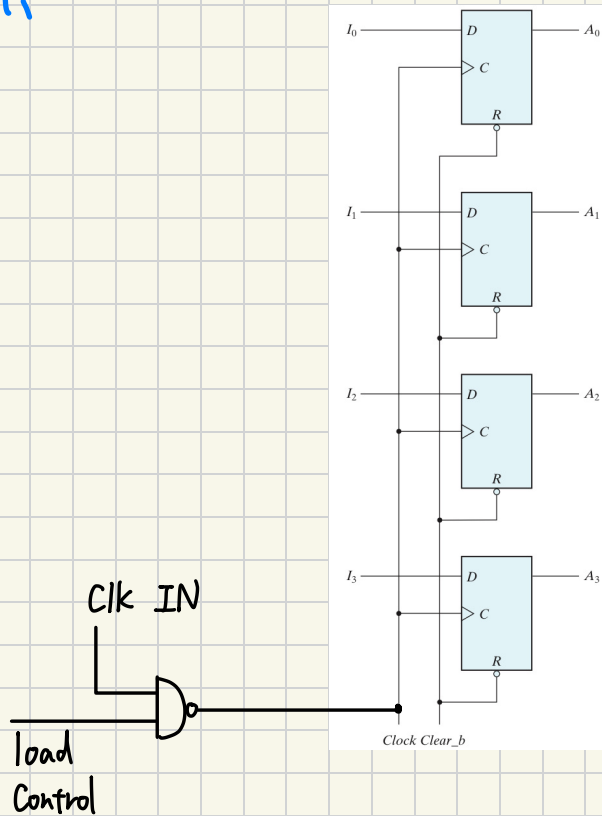
激勵表

$Q(t)$	$Q(t+1)$	J	K
0	0	0	x
0	1	0	x
1	0	x	1
1	1	x	0

$Q(t)$	$Q(t+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

Chapter 6.

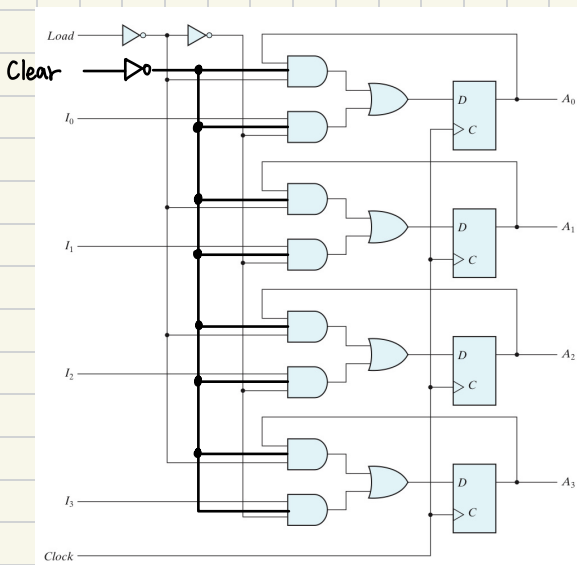
1、



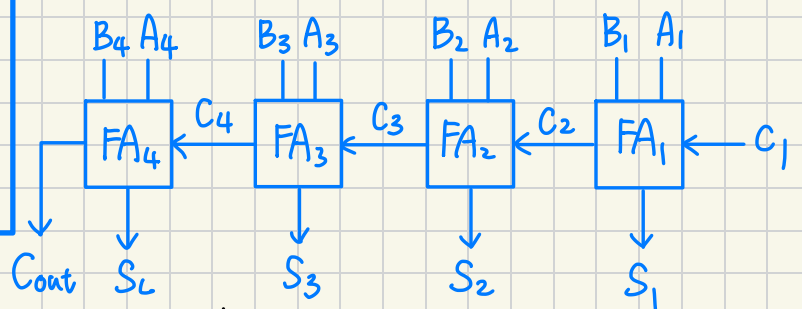
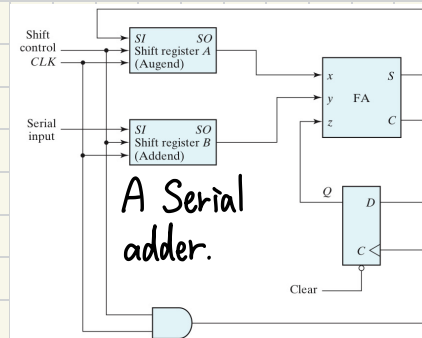
此电路使用 NAND 閘來控制 Clock，
假设 Load Control 並不是同步的，可能
导致以下幾個問題：

- 1、因 Clock 並不直接送給正反器，而先經過了 NAND 閘，這可能導致傳輸延遲、破壞電路的同步性。
- 2、閘控時脈可能會在正反器 Clock 的建立時間前到達，導致亞穩態。
- 3、Load Signal 將會截短時脈的訊號寬度。

2、



3、



A Parallel Adder.

Serial Adder 是一種使用了移位暫存器、全加器和一個正反器的序項邏輯電路。

Parallel Adder 則是僅利用全加器的組合電路。在這樣的情況下， n 位數的二進位數字進行加減，就需要用到 n 個全加器。

Parallel Adder 可以在 1 個 Clock Cycle 內輸出結果，Serial Adder 則要用 n 個 Clock Cycle。

Parallel Adder 較快，電路較複雜；Serial Adder 較慢，電路較簡單，不受數字位數限制。

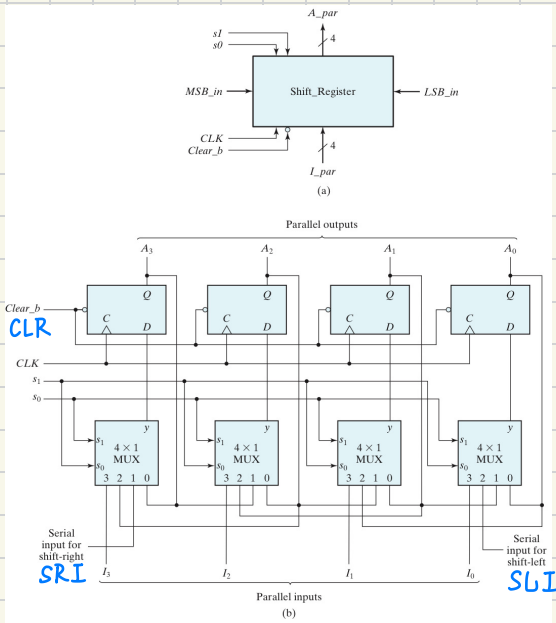
4.

Initial: 1010 , Input: 1011001

1010 → 1101 → 0110 → 0011 → 1001 → 1100 → 0110 → 1011

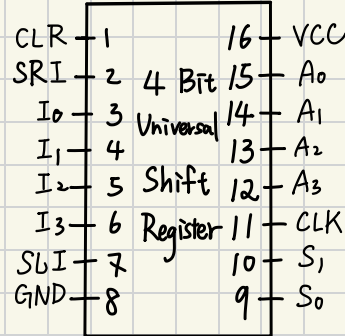
1 0 0 1 1 0 1

5.

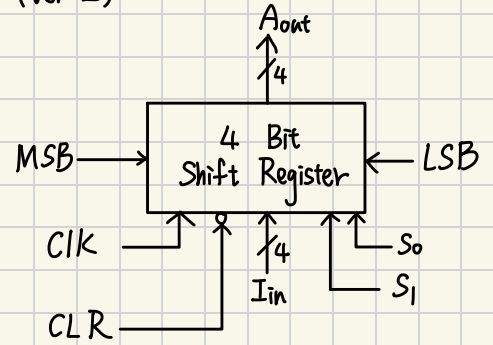


(a)

(Ver 1)

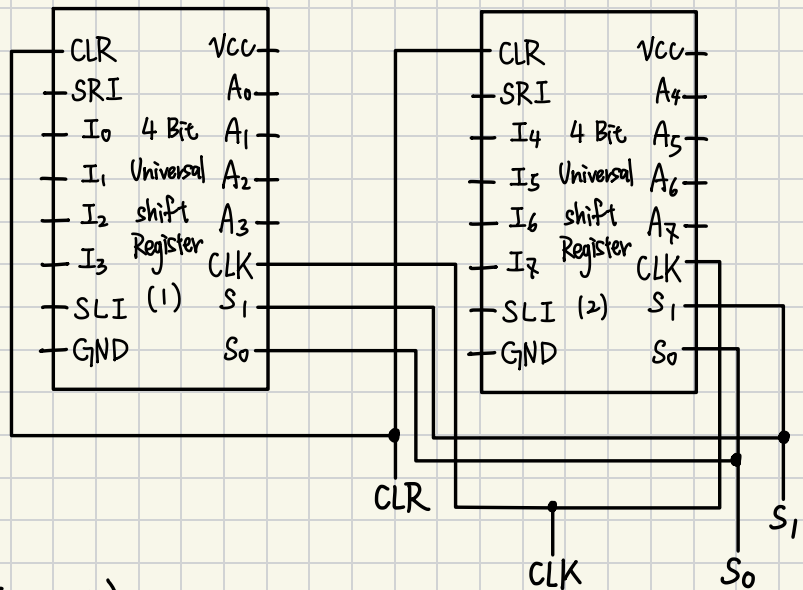


(Ver 2)

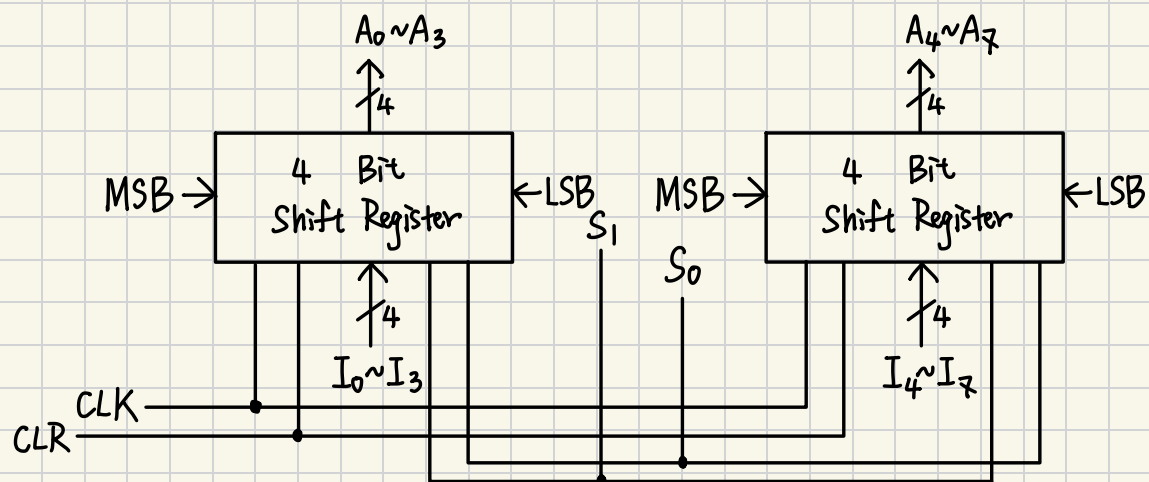


(b)

(Ver 1)

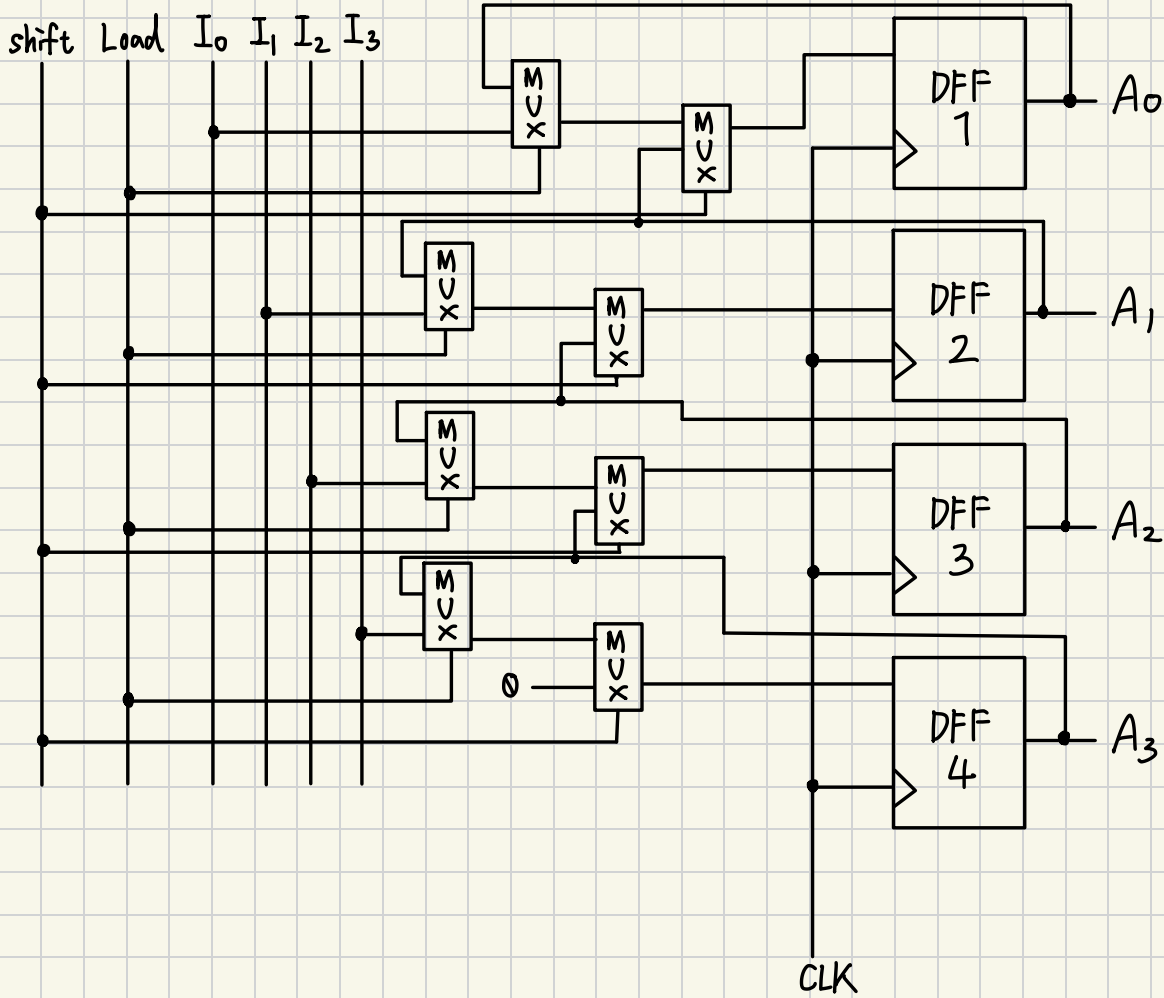
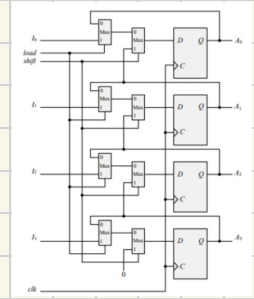


(Ver 2)



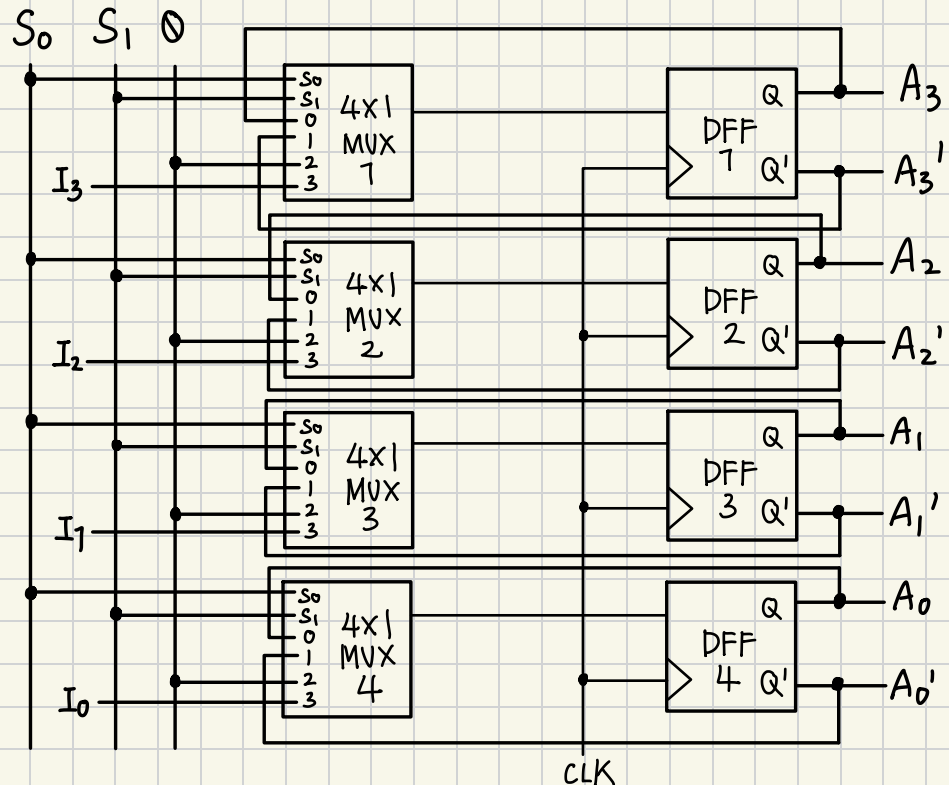
6.

load	shift	State
0	0	No Change
0	1	shift Right (to LSB)
1	0	Parallel Load
1	1	Shift Right



7.

S ₁	S ₀	State
0	0	No Change
0	1	Comp'
1	0	CLR
1	1	Parallel Load



8.

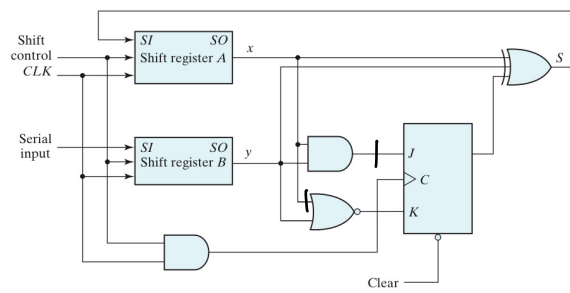


FIGURE 6.6
Second form of serial adder

J	K	State
0	0	NO CHANGE
0	1	0 RESET
1	0	1 SET
1	1	Comp!

Register A	Register B	Carry	Sum
0101	0111	0	0
0010	0011	1	0
0001	0001	1	0
1000	0000	1	1
1100	0000	0	1

Register A: 0101 → 0010 → 0001 → 1000 → 1100
Carry: 0 → 1 → 1 → 1 → 0

加到A的左
↓

9.

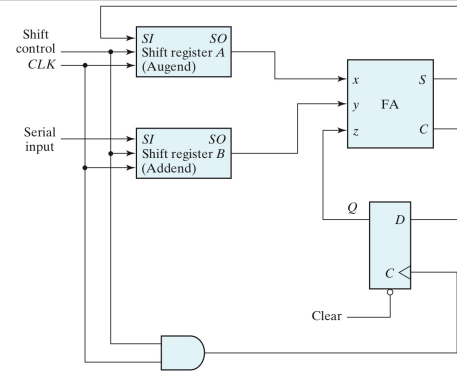


FIGURE 6.5
Serial adder

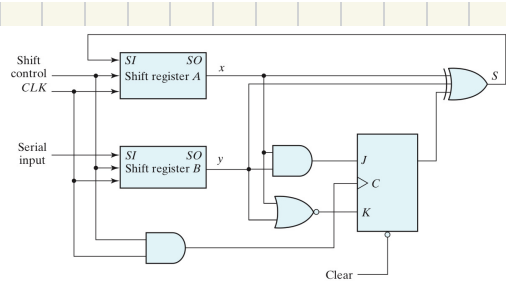


FIGURE 6.6
Second form of serial adder

(a)

* = 進位取補: 取NOT後+1

將 Serial Input 加上 NOT 取補, 並將 FA 的進位 C 設為 +1.

(b)

PS	IP	NS	OP	J	K	Q ^{xy}	00	01	11	10
Q	x y	Q	D			0	0	1	0	0
0	0 0	0	0	0	x	1	x	x	x	x
0	0 1	1	1	0	x					
0	1 0	0	0	0	x					
0	1 1	0	0	0	x					
1	0 0	1	1	x	0					
1	0 1	1	0	x	0					
1	1 0	0	0	x	1					
1	1 1	1	1	x	0					

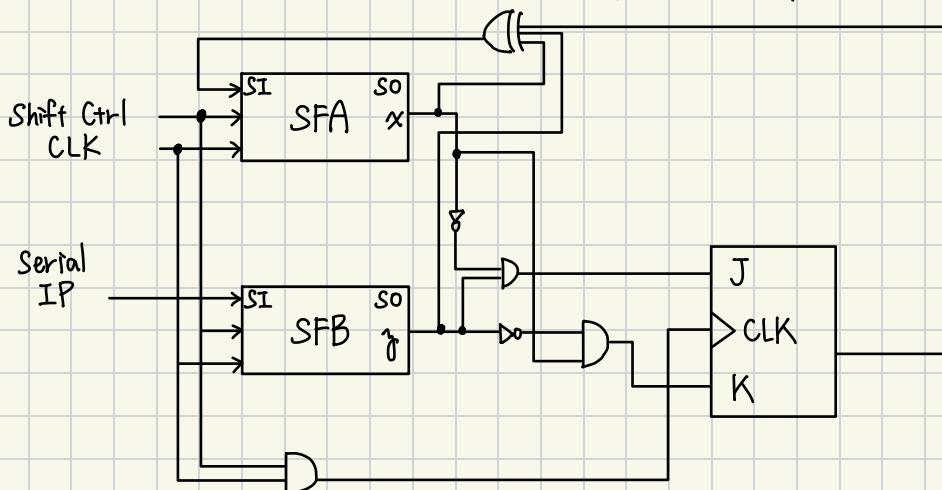
JK 激勵表

Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

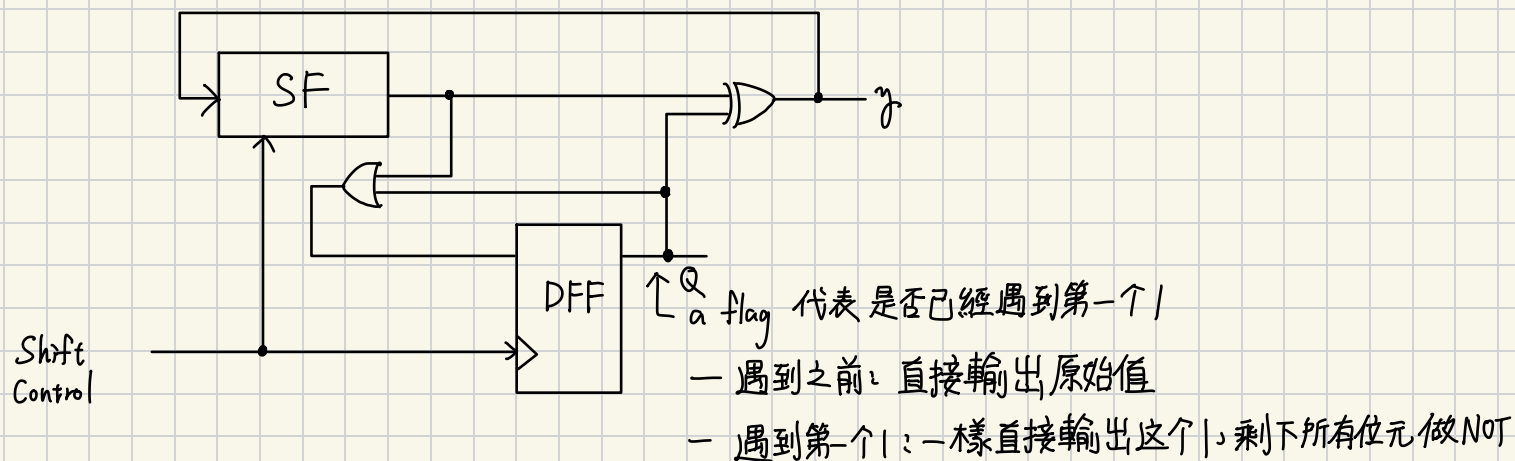
$$JK = JQ' + K'Q$$

Q ^{xy}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D = x \oplus y \oplus Q$$



10.



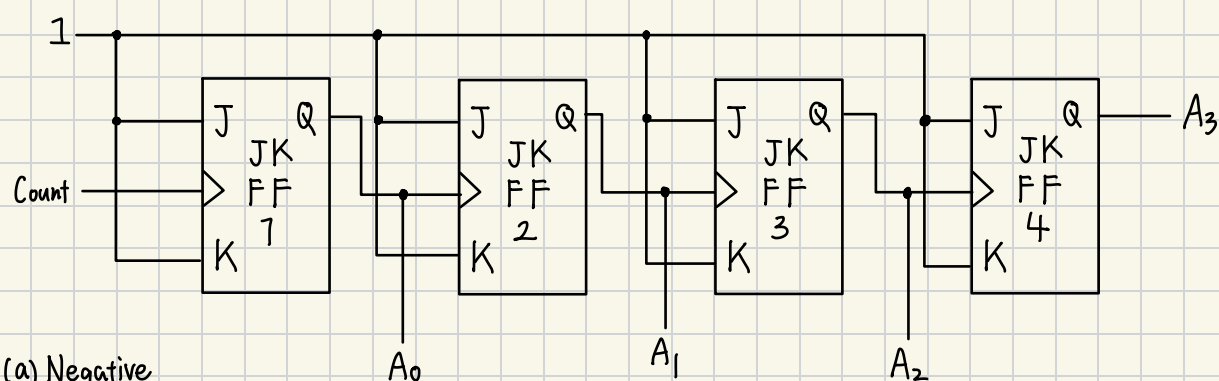
11.

<key: 負縁触発>

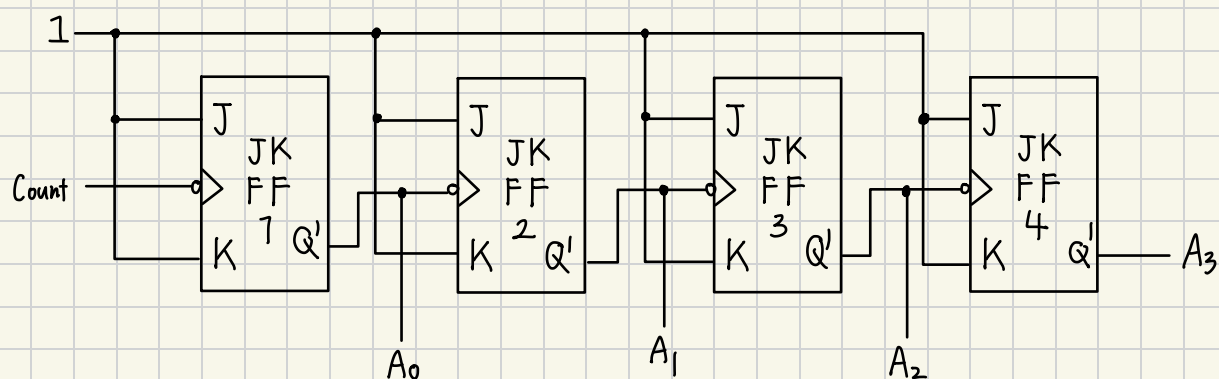
- a. Count up: 將各級正反器的反向輸出接到下一級正反器的輸入。
b. Count down: 將各級正反器的正向輸出接到下一級正反器的輸入。

72.

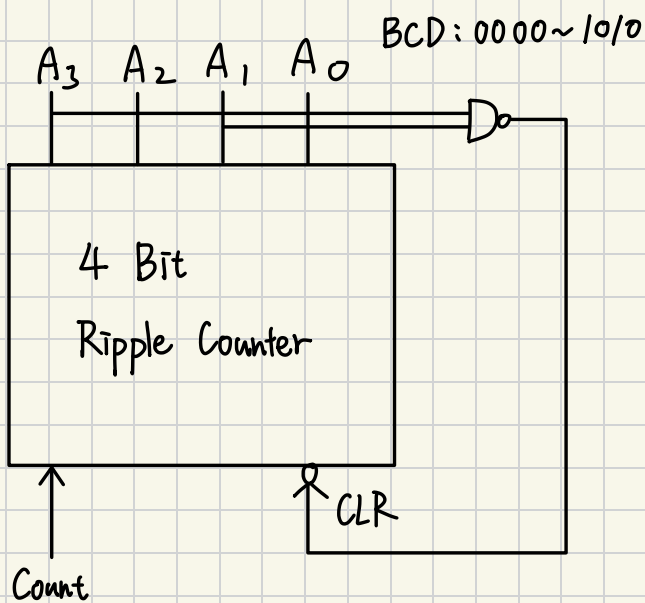
(a) Positive



(a) Negative



13



14

(a) 1100 1101 1011 : 3 个 Bit

(b) 0000 0011 1111 : 7 个 Bit

(c) 1110 1111 1111 : 9 个 Bit

15. 沒教，不会考

$$8 \times 4 = 32ns$$

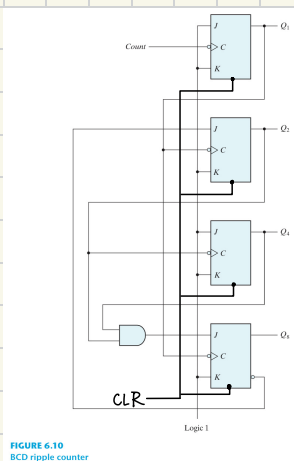
$$\frac{10^9}{32} = 31250000Hz = 31.25MHz$$

16

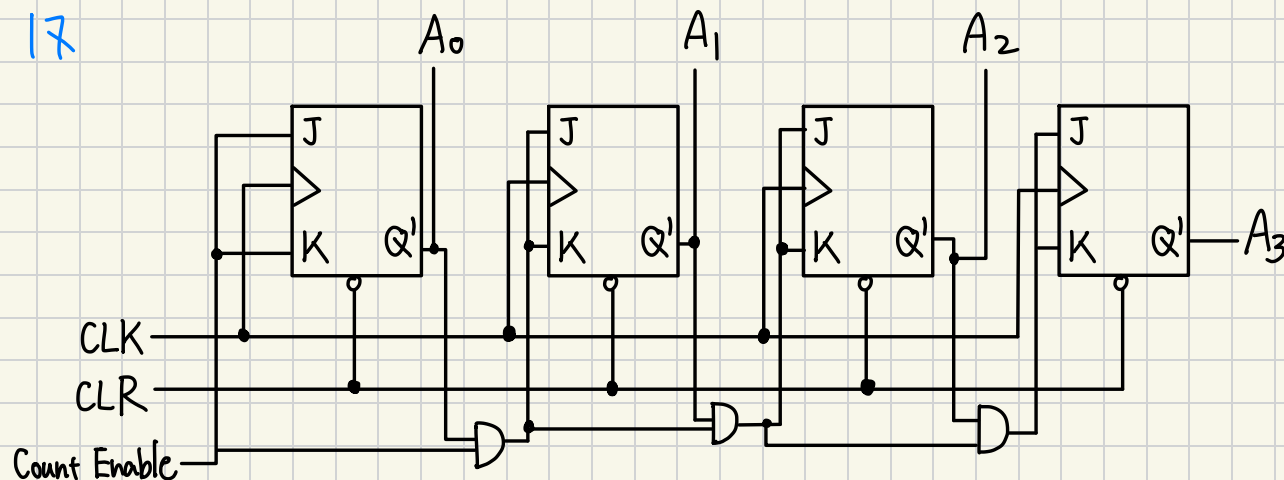
BCD: 0000 ~ 1001

Unused State	Next State	Next State
$Q_8 Q_4 Q_2 Q_1$	$Q_8 Q_4 Q_2 Q_1$	$Q_8 Q_4 Q_2 Q_1$
1 0 1 0	1 0 1 1	0 1 0 0
1 1 0 0	1 1 0 1	0 1 0 0
1 1 1 0	1 1 1 1	0 0 0 0

noise signal 若把 Counter 送入未使用態，每个未使用狀態最多在 2 个狀態後，也会自我恢復成合法狀態。



17



18、

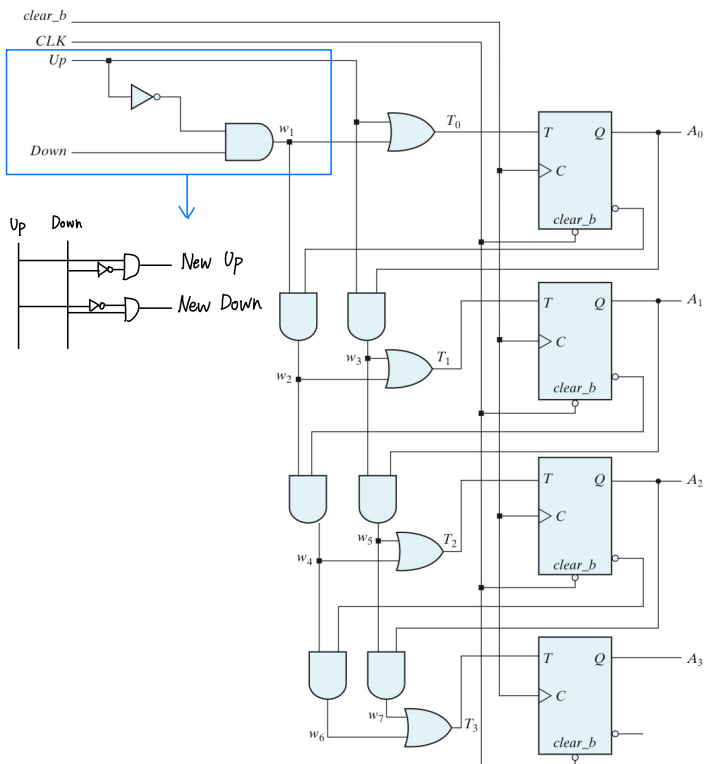


FIGURE 6.13 Four-bit up-down binary counter

When the up=down=1,
將會上數。

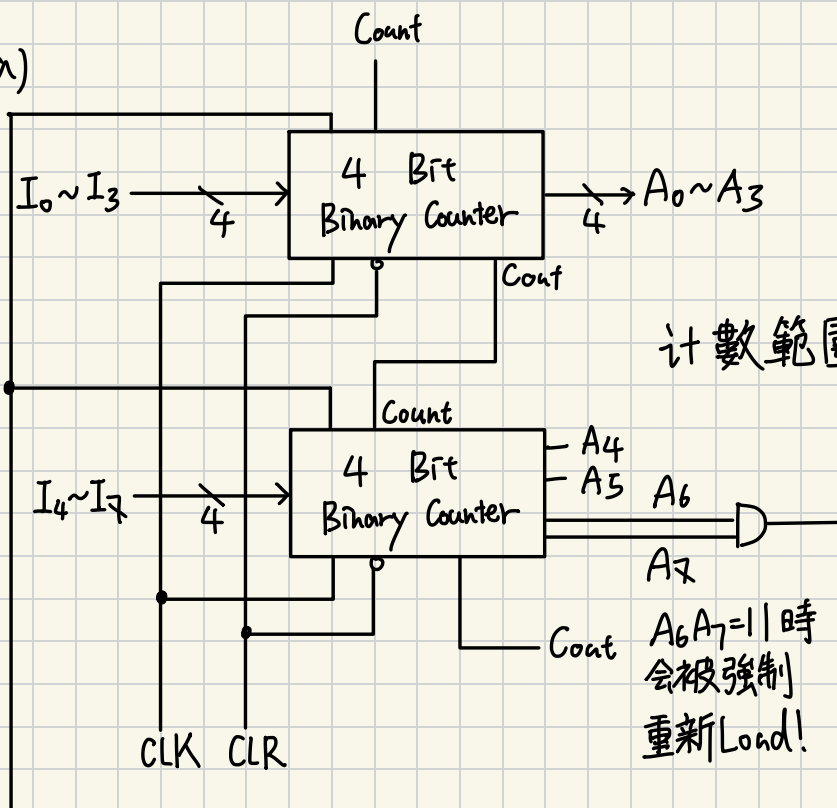
U	D	X	Y
0	0	0	0
0	1	0	1
1	0	1	0
1	1	0	0

U/D	0	1	U/D	0	1
0	0	0	0	0	1
1	1	0	1	0	0

UD' UD'

20、

(a)



計數範圍: 00000000 ~ 00111111

A6A7=11時
會被強制
重新Load!

